19. Which one of the following is not correct?

a) $La(OH)_3$ is less basic than $Lu(OH)_3$

b) In lanthanoid series ionic radius of Ln^{3+} ions decreases

c) La is actually an element of transition metal series rather than lanthanoid series

d) Atomic radii of Zr and Hf are same because of lanthanoid contraction

Answer the following questions:

1. What are transition metals? Give four examples.
2. Explain the oxidation states of 4d series elements.
3. What are inner transition elements?
4. Justify the position of lanthanoids and actinoids in the periodic table.
5. What are actinides? Give three examples.
6. Describe the preparation of potassium dichromate.
7. What is lanthanoid contraction and what are the effects of lanthanoid contraction?
8. Complete the following
   a. $MnO_4^{2-} + H^+ \longrightarrow ?$
   b. $C_6H_5CH_3 \xrightarrow[KMnO_4]{acidified} ?$
   c. $MnO_4^- + Fe^{2+} \longrightarrow ?$
   d. $KMnO_4 \xrightarrow[\text{Red hot}]{\Delta} ?$
   e. $Cr_2O_7^{2-} + I^- + H^+ \longrightarrow ?$
   f. $Na_2Cr_2O_7 + KCl \longrightarrow ?$
9. What are interstitial compounds?
10. Calculate the number of unpaired electrons in Ti^{3+} , Mn^{2+} and calculate the spin only magnetic moment.
11. Write the electronic configuration of Ce^{4+} and Co^{2+}.
12. Explain briefly how +2 states becomes more and more stable in the first half of the first row transition elements with increasing atomic number.
13. Which is more stable? Fe^{3+} or Fe^{2+} - explain.
14. Explain the variation in $E^0_{M^{3+}/M^{2+}}$ 3d series.

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